



ACCESS S3 FROM A VPC

By Nabil Ahmed | September 2026

Access S3 from a VPC

```
[ec2-user@ip-10-0-10-29 ~]$ sudo touch /tmp/test.txt
[ec2-user@ip-10-0-10-29 ~]$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-nabil
upload: ../../tmp/test.txt to s3://nextwork-vpc-project-nabil/test.txt
[ec2-user@ip-10-0-10-29 ~]$ aws s3 ls s3://nextwork-vpc-project-nabil
2026-09-08 14:37:43      637237 legendary-aws-networks-monitoring.pdf
2026-09-08 14:37:43      726884 legendary-aws-networks-peering.pdf
2026-09-08 14:48:11           0 test.txt
```

Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is AWS's foundational networking service that allows us to create our own isolated networks within an AWS region and control network traffic and security etc.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to launch an EC2 instance inside a public subnet and directly accessed/managed an Amazon S3 bucket through the EC2 instance using the AWS CLI.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was learning that using access keys are not the best practice method despite being the most simple to enable a resource inside a public subnet to be able to manage resources in your own AWS environment.

This project took me...

This project took me one hour to complete including documentation.

In the first part of my project...

Step 1 - Architecture set up

In this step, I will I will launch a VPC with a public subnet. I will also launch an EC2 instance inside the public subnet.

Step 2 - Connect to my EC2 instance

In this step, I will directly access an EC2 instance using EC2 Instance Connect.

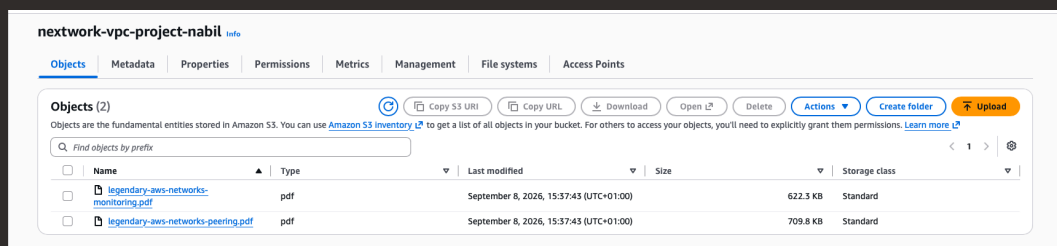
Step 3 - Set up access keys

In this step, I will create access keys so that my EC2 instance can have access to my AWS environment, specifically, the ability to interact with an S3 bucket.

Architecture set up

I started my project by launching a VPC with a public subnet, and an EC2 instance inside the public subnet.

I also set up an S3 bucket with two files inside.



Running CLI commands

AWS CLI is a software we can download into a local computer's terminal so that we can have access to our own AWS account and different actions without needing to use the AWS Management Console, I have access to AWS CLI because it is pre-installed in EC2 instances.

The first command I ran was 'aws s3 ls'. This command is used to list all the S3 buckets inside the AWS account (the buckets that the EC2 instances/application has access to).

The second command I ran was 'aws configure'. This command is used to set up my EC2 instance's credentials in order to access my AWS environment.

```

      #
    ~- \##### Amazon Linux 2023
    ~- \#####
    ~- \###|
    ~- \#|
    ~- v- ' ' -> https://aws.amazon.com/linux/amazon-linux-2023
    ~-
    ~- .
    ~- /m/ ' '
[ec2-user@ip-10-0-10-29 ~]$ aws s3 ls

Unable to locate credentials. You can configure credentials by running "aws login".
[ec2-user@ip-10-0-10-29 ~]$ aws configure
AWS Access Key ID [None]:

```

Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, I configured an Access Key ID, Secret Access Key, default region and default output format.

Access keys are credentials that my EC2 instance and other applications/servers need to use in order to get access to my AWS environment i.e. interact with my AWS resources and services.

Secret access keys are like passwords for our access keys/credentials. My EC2 instances/other applications would need secret access keys as authentication and log in to my AWS environment.

Best practice

Although I'm using access keys in this project, a best practice alternative is to use IAM roles with permissions attached. This is a more secure way to grant access to an EC2 instance because it is much easier to track, attach and detach IAM policies.

In the second part of my project...

Step 4 - Set up an S3 bucket

In this step, I will launch an Amazon S3 bucket with two files inside. This S3 bucket will be accessed by my EC2 instance later on in the project, so I can test whether my access key has successfully given AWS access to my EC2 instance.

Step 5 - Connecting to my S3 bucket

In this step, I am using AWS CLI commands to try control/manage my S3 bucket. This means I am interacting with my S3 bucket through my EC2 instance/VPC instead of the AWS Management Console.

Connecting to my S3 bucket

The first command I ran was 'aws s3 ls'. This command is used to list all the S3 buckets inside the AWS account (the buckets that the EC2 instances/application has access to).

When I ran the command 'aws s3 ls' again, the terminal responded with a list of my S3 buckets. This indicated that my access key works i.e. my EC2 instance has access to my AWS environment.

```
[ec2-user@ip-10-0-10-29 ~]$ aws s3 ls
2026-09-08 14:35:52 nextwork_vpc-project-nabil
```

Connecting to my S3 bucket

Another CLI command I ran was 'aws s3 ls s3://nextwork-vpc-project-nabil' which returned the files that were stored inside my S3 bucket.

```
[ec2-user@ip-10-0-10-29 ~]$ aws s3 ls s3://nextwork-vpc-project-nabil
2026-09-08 14:37:43      637237 legendary-aws-networks-monitoring.pdf
2026-09-08 14:37:43      726884 legendary-aws-networks-peering.pdf
```

Uploading objects to S3

To upload a new file to my bucket, I first ran the command 'sudo touch /tmp/test.txt'. This command creates a file called test.txt inside the temp directory.

The second command I ran was. 'aws s3 cp /tmp/test.txt s3://network-vpc-project-nabil'. This command will upload the test.txt file that I just created into my S3 bucket.

The third command I ran was 'aws s3 ls s3://network-vpc-project-nabil' which validated that the test.txt file was uploaded into the S3 bucket.

```
[ec2-user@ip-10-0-10-29 ~]$ sudo touch /tmp/test.txt
[ec2-user@ip-10-0-10-29 ~]$ aws s3 cp /tmp/test.txt s3://network-vpc-project-nabil
upload: ../../tmp/test.txt to s3://network-vpc-project-nabil/test.txt
[ec2-user@ip-10-0-10-29 ~]$ aws s3 ls s3://network-vpc-project-nabil
2026-09-08 14:37:43      637237 legendary-aws-networks-monitoring.pdf
2026-09-08 14:37:43      726884 legendary-aws-networks-peering.pdf
2026-09-08 14:48:11           0 test.txt
```